1. Friction is a tangential resistive force that develops at the contact interface of two bodies to oppose:

(A) Impending or actual relative sliding motion
(B) Pure normal contact compressive pressure
(C) Gravitational mass acceleration
(D) Rotational angular inertia

2. Frictional resistance existing between surfaces prior to the onset of relative sliding is classified as:

(A) Static Friction
(B) Kinetic Friction
(C) Rolling Resistance
(D) Viscous Drag

3. The maximum tangential frictional resistance developed just as sliding is on the verge of occurring is called:

(A) Limiting Friction
(B) Residual Friction
(C) Dynamic Drag
(D) Hydrodynamic Friction

4. Static friction F_s is characterized physically as a:

(A) Self-adjusting force from zero up to limiting friction
(B) Constant invariant force independent of applied load
(C) Negative conservative potential field
(D) Velocity-dependent force

5. Frictional resistance opposing relative motion during actual ongoing macroscopic sliding is termed:

(A) Kinetic (Dynamic) Friction
(B) Limiting Friction
(C) Static Threshold
(D) Molecular Stiction

6. For any given pair of dry engineering contacting surfaces, the relationship between coefficients of friction is:

(A) mu_k is strictly less than mu_s (mu_k < mu_s)
(B) mu_k is always greater than mu_s
(C) mu_k is identically equal to mu_s
(D) mu_k is infinite in static state

7. According to Coulomb's first law of dry friction, limiting friction F_lim is directly proportional to:

(A) The normal reaction force N
(B) The macroscopic apparent contact surface area
(C) The sliding velocity of the body
(D) The ambient atmospheric pressure

8. Coulomb's second law of dry friction establishes that limiting friction force is strictly independent of:

(A) The apparent macroscopic area of contact
(B) The chemical nature of contacting materials
(C) The magnitude of normal contact reaction
(D) The surface roughness texture

9. Coulomb's third law states that kinetic friction over moderate sliding velocity ranges is:

(A) Virtually independent of the relative velocity of sliding
(B) Directly proportional to the cube of velocity
(C) Exponentially decaying with time
(D) Inversely proportional to displacement

10. The dimensionless coefficient of static friction mu_s is mathematically defined as the ratio:

(A) F_lim / N
(B) N / F_lim
(C) F_lim * N
(D) sqrt(F_lim / N)

11. The angle between the resultant contact reaction R and the normal reaction N at impending slip is the:

(A) Angle of Limiting Friction (phi_s)
(B) Angle of Repose
(C) Helix Pitch Angle
(D) Semi-Apex Cone Angle

12. The mathematical relationship connecting the coefficient of static friction mu_s and angle of friction phi_s is:

(A) mu_s = tan(phi_s)
(B) mu_s = sin(phi_s)
(C) mu_s = cos(phi_s)
(D) mu_s = cot(phi_s)

13. The maximum angle of an inclined plane at which a granular or solid body rests in static equilibrium without sliding is the:

(A) Angle of Repose (theta_r)
(B) Angle of Wedge Taper
(C) Angle of Wrap
(D) Lead Angle of Screw

14. For a body resting freely on an inclined plane under gravity alone, the angle of repose theta_r is identically equal to:

(A) The angle of static friction (theta_r = phi_s)
(B) Twice the angle of friction
(C) Half the angle of friction
(D) 90 degrees minus phi_s

15. The geometrical right circular cone generated by rotating the resultant contact reaction R around the normal axis is the:

(A) Cone of Friction
(B) Mohr Failure Envelope
(C) Poinsot Central Cone
(D) Coulomb Critical Cone

16. The semi-vertical apex angle of the Cone of Friction is equal to:

(A) The angle of static friction phi_s
(B) Twice the angle of friction (2 * phi_s)
(C) The angle of repose minus 45 degrees
(D) The helix angle of the thread

17. If the total resultant contact force vector R lies strictly inside the cone of friction, the contacting body:

(A) Remains in stable static equilibrium without sliding
(B) Is on the verge of impending motion
(C) Undergoes accelerated sliding motion
(D) Experiences mechanical overturning

18. If the resultant contact force vector R lies on the outer boundary surface of the cone of friction:

(A) The body is in a state of impending slip (limiting equilibrium)
(B) The body is floating freely
(C) The coefficient of friction drops to zero
(D) The body is moving at supersonic speed

19. A wedge is a simple mechanical machine utilized in civil and geotechnical engineering primarily to:

(A) Lift heavy loads or split blocks through lateral driving forces
(B) Eliminate all normal contact stresses
(C) Transform rotational motion into harmonic wave energy
(D) Store kinetic energy as a flywheel

20. In wedge mechanics, the direction of the frictional resistance force acting on each contact interface is always:

(A) Opposite to the direction of impending relative sliding
(B) Parallel to the applied driving force
(C) Perpendicular to the contact interface
(D) Directly collinear with the normal force

21. A power screw mechanism utilized for lifting vertical compressive loads through applied torque is called a:

(A) Screw Jack
(B) Turnbuckle
(C) Worm Gearbox
(D) Differential Pulley

22. The linear axial distance advanced by a screw in one complete 360-degree revolution is termed the:

(A) Pitch (p) or Lead
(B) Helix Angle
(C) Mean Diameter
(D) Core Root Width

23. For a single-start power screw thread, the lead L is equal to:

(A) The pitch p (L = p)
(B) Twice the pitch (L = 2*p)
(C) Half the pitch (L = 0.5*p)
(D) pi times the pitch

24. The helix lead angle alpha of a screw thread having pitch p and mean diameter d_m is calculated from:

(A) tan(alpha) = p / (pi * d_m)
(B) sin(alpha) = p / (2 * pi * d_m)
(C) tan(alpha) = (pi * d_m) / p
(D) cos(alpha) = p / d_m

25. In a square-threaded screw jack lifting a vertical load W, the tangential effort P required at the mean radius is:

(A) P = W * tan(alpha + phi)
(B) P = W * tan(alpha - phi)
(C) P = W * sin(alpha + phi)
(D) P = W / tan(alpha + phi)

26. In a square-threaded screw jack lowering a vertical load W, the tangential effort P required at the mean radius is:

(A) P = W * tan(phi - alpha)
(B) P = W * tan(phi + alpha)
(C) P = W * cot(phi - alpha)
(D) P = W * cos(phi + alpha)

27. A screw jack is defined as 'self-locking' if the load will not lower under its own weight when the driving effort is removed. The condition is:

(A) alpha <= phi (Helix angle <= Angle of friction)
(B) alpha > phi
(C) alpha = 45 degrees
(D) mu = 0

28. If the helix angle alpha exceeds the angle of friction phi (alpha > phi), the screw jack is classified as:

(A) Overhauling (Reversible)
(B) Self-locking
(C) Statically Indeterminate
(D) Kinematically Locked

29. The mechanical efficiency eta of a square-threaded screw jack lifting a load W is given by:

(A) eta = tan(alpha) / tan(alpha + phi)
(B) eta = tan(alpha + phi) / tan(alpha)
(C) eta = sin(alpha) * cos(phi)
(D) eta = 1 - tan(alpha)

30. The maximum theoretical efficiency of a self-locking square-threaded screw jack (where alpha <= phi) is strictly less than:

(A) 50%
(B) 100%
(C) 75%
(D) 25%

31. The exponential governing equation for belt friction over a fixed curved drum (Capstan equation) relates tight tension T1 and slack tension T2 by:

(A) T1 / T2 = exp(mu * theta)
(B) T1 / T2 = mu * theta
(C) T1 / T2 = log(mu * theta)
(D) T1 - T2 = exp(mu * theta)

32. In the Capstan belt friction equation T1 / T2 = exp(mu * theta), the contact lap angle theta must strictly be expressed in:

(A) Radians
(B) Degrees
(C) Gradians
(D) Revolutions

33. The net effective driving tangential force transmitted by a belt drive is:

(A) T1 - T2
(B) T1 + T2
(C) T1 * T2
(D) sqrt(T1^2 + T2^2)

34. Power P transmitted by a belt running at linear speed v with tensions T1 and T2 is calculated as:

(A) P = (T1 - T2) * v
(B) P = (T1 + T2) * v
(C) P = (T1 - T2) / v
(D) P = 0.5 * (T1 - T2) * v^2

35. For a V-belt running in a grooved pulley with included groove angle 2*beta, the effective coefficient of friction mu_prime is:

(A) mu / sin(beta)
(B) mu * sin(beta)
(C) mu / cos(beta)
(D) mu * tan(beta)

36. Centrifugal tension T_c developed in a high-speed belt having mass m per unit length moving at linear velocity v is:

(A) T_c = m * v^2
(B) T_c = 0.5 * m * v^2
(C) T_c = m * v
(D) T_c = m * g * v

37. For maximum power transmission in a flexible belt drive, the centrifugal tension T_c should equal what fraction of maximum allowable tension T_max?

(A) T_c = T_max / 3
(B) T_c = T_max / 2
(C) T_c = T_max / 4
(D) T_c = 2 * T_max / 3

38. Rolling resistance opposes the rolling motion of a wheel primarily due to:

(A) Inelastic viscoelastic deformation and hysteresis of contacting materials
(B) Coulomb microscopic sliding adhesion
(C) Aerodynamic boundary layer skin friction
(D) Gyroscopic precession torques

39. The coefficient of rolling resistance c_r (or a) has the physical dimension of:

(A) Length (metres)
(B) Dimensionless
(C) Force (Newtons)
(D) Velocity (m/s)

40. In ladder friction problems, when the ladder base rests on a rough horizontal floor and leans against a smooth vertical wall, the reaction at the wall is:

(A) Purely normal (horizontal)
(B) Purely vertical (upward)
(C) Inclined at 45 degrees
(D) Zero

41. A block of weight W = 500 N rests on a horizontal floor with mu_s = 0.35. The maximum horizontal force that can be applied without causing motion is:

(A) 175 N
(B) 143 N
(C) 500 N
(D) 250 N

42. If a horizontal force of 100 N is applied to the 500 N block in Q41 (mu_s = 0.35), the actual static friction force developed is:

(A) 100 N (Self-adjusting static friction)
(B) 175 N
(C) 0 N
(D) 75 N

43. A 200 N block rests on a rough inclined plane of 30 degrees. If mu_s = 0.40, the component of gravity down the plane is:

(A) 100 N
(B) 173.2 N
(C) 200 N
(D) 69.3 N

44. In Q43 (W = 200 N, slope 30 deg, mu_s = 0.40), the normal reaction N exerted by the plane is:

(A) 173.2 N
(B) 100 N
(C) 200 N
(D) 86.6 N

45. In Q43-44, the maximum available limiting friction force up the plane is:

(A) 69.28 N
(B) 100 N
(C) 173.2 N
(D) 40 N

46. Since the gravity component down the plane (100 N) exceeds limiting friction (69.28 N) in Q43-45, the block will:

(A) Slide down the plane with accelerated motion
(B) Remain in stable static equilibrium
(C) Move up the plane
(D) Topple over its corner

47. The minimum force P acting parallel to the 30-degree incline required to prevent the 200 N block in Q43-46 from sliding down is:

(A) 30.72 N
(B) 69.28 N
(C) 100.00 N
(D) 169.28 N

48. The minimum force P parallel to the incline required to pull the 200 N block up the 30-degree plane in Q43-46 is:

(A) 169.28 N
(B) 100.00 N
(C) 69.28 N
(D) 30.72 N

49. A block of weight 1000 N rests on a horizontal surface (mu_s = 0.25). The most efficient pull force angle theta with the horizontal to minimize required force is:

(A) theta = phi_s = atan(0.25) = 14.04 degrees
(B) theta = 0 degrees
(C) theta = 45 degrees
(D) theta = 30 degrees

50. In Q49, the minimum magnitude of the pulling force P applied at optimal angle theta = 14.04 deg is:

(A) 242.54 N
(B) 250.00 N
(C) 300.00 N
(D) 200.00 N

51. A 5 m uniform ladder weighing 300 N leans at 60 deg to the horizontal against a smooth wall. If the floor is rough (mu_s = 0.3), the limiting friction at the floor is:

(A) 90 N
(B) 150 N
(C) 300 N
(D) 86.6 N

52. In Q51, the normal reaction exerted by the smooth vertical wall on the ladder top is:

(A) 86.60 N
(B) 90.00 N
(C) 150.00 N
(D) 300.00 N

53. Since the required horizontal thrust at the floor (86.60 N) is less than the limiting friction (90 N) in Q51-52, the ladder:

(A) Remains in stable static equilibrium without slipping
(B) Slips immediately
(C) Lifts off the floor
(D) Rotates clockwise

54. A uniform ladder leans against a smooth wall and rough floor (mu_s). The critical minimum angle theta_min with the horizontal for equilibrium under self-weight is:

(A) tan(theta_min) = 1 / (2 * mu_s)
(B) tan(theta_min) = 1 / mu_s
(C) tan(theta_min) = 2 * mu_s
(D) theta_min = phi_s

55. If mu_s = 0.40 between floor and ladder, the minimum safe angle of the ladder with the horizontal under its own weight is:

(A) 51.34 degrees
(B) 38.66 degrees
(C) 60.00 degrees
(D) 45.00 degrees

56. A 10-degree wedge is driven under a 5000 N vertical column. If friction coefficient mu = 0.25 on all surfaces (phi = 14.04 deg), the driving force P is:

(A) 2648.7 N
(B) 1250.0 N
(C) 5000.0 N
(D) 882.5 N

57. In Q56, the condition for the wedge to be self-locking after removal of driving force P is:

(A) alpha <= 2 * phi
(B) alpha > 2 * phi
(C) alpha = 90 degrees
(D) mu = 0

58. Since alpha = 10 deg and 2 * phi = 28.08 deg in Q56-57, the wedge:

(A) Is fully self-locking (will not pop out)
(B) Will slip backward automatically
(C) Has zero holding capacity
(D) Requires a negative pull force

59. A square-threaded screw jack has mean diameter d_m = 50 mm and pitch p = 10 mm. Its thread lead angle alpha is:

(A) 3.64 degrees
(B) 7.28 degrees
(C) 1.82 degrees
(D) 11.31 degrees

60. If mu = 0.15 for the screw jack in Q59 (phi = 8.53 deg), the angle (alpha + phi) during lifting is:

(A) 12.17 degrees
(B) 8.53 degrees
(C) 3.64 degrees
(D) 15.81 degrees

61. For the screw jack in Q59-60 lifting a load W = 20 kN, the tangential effort P required at the mean radius is:

(A) 4.31 kN
(B) 3.00 kN
(C) 1.27 kN
(D) 20.00 kN

62. If a 0.5 m long tommy bar lever is used to turn the screw jack in Q59-61, the manual force F_bar required at the lever tip is:

(A) 215.5 N
(B) 431.0 N
(C) 107.8 N
(D) 862.0 N

63. The mechanical efficiency of the screw jack in Q59-61 is:

(A) 29.74%
(B) 45.00%
(C) 60.50%
(D) 15.20%

64. A flat belt wraps around a fixed circular bollard with 1.5 complete turns. The total angle of wrap theta in radians is:

(A) 3 * pi radians (9.425 rad)
(B) 1.5 * pi radians
(C) 6 * pi radians
(D) pi radians

65. If mu = 0.30 and a sailor exerts a slack pull T2 = 50 N on the bollard in Q64 (theta = 3*pi rad), the holding tension T1 is:

(A) 843.8 N
(B) 150.0 N
(C) 50.0 N
(D) 2500.0 N

66. If the belt wraps around the bollard for 3 complete turns (theta = 6*pi rad, mu = 0.3), the tension ratio T1 / T2 becomes:

(A) 285.4
(B) 16.9
(C) 6.28
(D) 1000.0

67. A flat belt drives a pulley of diameter 1.2 m at 300 rpm. The linear belt speed v is:

(A) 18.85 m/s
(B) 37.70 m/s
(C) 9.42 m/s
(D) 60.00 m/s

68. In Q67, if tight tension T1 = 2500 N and slack tension T2 = 1000 N, the power transmitted by the belt is:

(A) 28.27 kW
(B) 15.00 kW
(C) 47.12 kW
(D) 9.42 kW

69. A belt has mass m = 0.8 kg/m and runs at speed v = 20 m/s. The centrifugal tension T_c developed in the belt is:

(A) 320 N
(B) 160 N
(C) 400 N
(D) 640 N

70. If maximum allowable belt tension T_max = 1800 N, the maximum permissible tight-side tension T1 (accounting for T_c = 320 N) is:

(A) 1480 N
(B) 1800 N
(C) 2120 N
(D) 1160 N

71. The optimum belt speed v_opt for transmitting maximum power when maximum tension T_max = 2400 N and mass m = 1.2 kg/m is:

(A) 25.82 m/s
(B) 44.72 m/s
(C) 15.00 m/s
(D) 35.50 m/s

72. A steel cylinder of weight 600 N and radius 0.2 m is pushed along a horizontal road with rolling resistance coefficient c_r = 0.005 m. The horizontal rolling resistance force is:

(A) 15 N
(B) 30 N
(C) 3 N
(D) 60 N

73. Two blocks A (weight 200 N) and B (weight 300 N) are placed in contact on a horizontal surface with mu_A = 0.2 and mu_B = 0.3. The horizontal force to cause impending slip of both is:

(A) 130 N
(B) 100 N
(C) 150 N
(D) 250 N

74. A 100 kg block is pulled up a 20-degree incline by a winch cable. If mu_k = 0.25, the required steady pulling tension is:

(A) 565.6 N
(B) 335.5 N
(C) 981.0 N
(D) 230.1 N

75. When an automobile brakes and skids on wet asphalt (mu_k = 0.5), its deceleration is:

(A) 4.905 m/s^2
(B) 9.81 m/s^2
(C) 2.45 m/s^2
(D) 0.5 m/s^2

76. A 12 m ladder (weight W = 400 N) leans against a rough wall (mu_w = 0.25) and rough floor (mu_f = 0.35) at 60 deg. A 800 N person climbs up. The max safe climbing distance x from base is:

(A) 7.88 m
(B) 6.00 m
(C) 10.50 m
(D) 4.25 m

77. For the ladder in Q76 at impending slip, the vertical reaction R_fy at the floor is:

(A) 1056.2 N
(B) 1200.0 N
(C) 950.0 N
(D) 1140.5 N

78. For the ladder in Q76-77, the horizontal reaction R_wx at the vertical wall is:

(A) 369.7 N
(B) 420.0 N
(C) 280.5 N
(D) 500.0 N

79. Two identical 8-degree wedges are used to raise a 12 kN heavy machine load vertically. Friction coefficient mu = 0.28 on all surfaces (phi = 15.64 deg). The driving force P on each wedge is:

(A) 7.78 kN
(B) 3.36 kN
(C) 12.00 kN
(D) 5.12 kN

80. In Q79, the horizontal force required to pull out the wedge (lowering the machine) is:

(A) - 3.14 kN (Pull force required, self-locking)
(B) + 3.14 kN
(C) 0 kN
(D) - 12.00 kN

81. A double-threaded square power screw has nominal diameter 60 mm, pitch 12 mm (lead L = 24 mm). The lead angle alpha is:

(A) 7.93 degrees
(B) 3.98 degrees
(C) 15.86 degrees
(D) 11.50 degrees

82. If mu = 0.12 (phi = 6.84 deg) for the double-threaded screw in Q81, the screw is:

(A) Overhauling (Since alpha = 7.93 deg > phi = 6.84 deg)
(B) Self-locking
(C) Statically Indeterminate
(D) Bound

83. For the screw in Q81-82 lowering an axial load W = 50 kN, the torque required to prevent spontaneous overhauling is:

(A) 27.6 N*m
(B) 55.2 N*m
(C) 110.0 N*m
(D) 0 N*m

84. In Q81-83, the lifting torque required to raise the 50 kN load with collar friction (mean collar diameter 90 mm, mu_c = 0.10) is:

(A) 627.5 N*m
(B) 402.5 N*m
(C) 225.0 N*m
(D) 850.0 N*m

85. A V-belt drives a dynamo transmitting 45 kW at linear speed v = 25 m/s. The groove angle 2*beta = 36 deg (beta = 18 deg). If mu = 0.3, the effective coefficient mu_prime is:

(A) 0.9708
(B) 0.3000
(C) 0.5000
(D) 0.1500

86. In Q85, if lap angle theta = 160 deg (2.793 rad), the tension ratio T1 / T2 is:

(A) 15.05
(B) 4.25
(C) 2.71
(D) 8.50

87. In Q85-86, the effective drive force (T1 - T2) is:

(A) 1800 N
(B) 45000 N
(C) 1125 N
(D) 2500 N

88. In Q85-87, the tight tension T1 in the belt is:

(A) 1928 N
(B) 1800 N
(C) 2500 N
(D) 3000 N

89. A differential band brake has drum diameter D = 600 mm, coefficient of friction mu = 0.3, and lap angle theta = 240 deg (4.189 rad). The tension ratio T1 / T2 is:

(A) 3.514
(B) 2.150
(C) 5.200
(D) 1.750

90. In Q89, if brake lever arms are a = 100 mm and b = 30 mm, the condition for self-locking in differential braking is:

(A) b / a >= 1 / (T1 / T2) = 0.2846
(B) b / a < 0.2846
(C) a = b
(D) mu = 0

91. A 2000 kg vehicle ascends a 15-degree slope at constant speed. If rolling resistance coefficient c_r = 0.02 and air drag is 250 N, total tractive force required is:

(A) 5704 N
(B) 5078 N
(C) 2500 N
(D) 6250 N

92. A heavy uniform rectangular crate of mass 200 kg (width 0.8 m, height 1.6 m) rests on a rough floor (mu_s = 0.4). A horizontal pull P is applied at height h. The max height h for slip before tipping is:

(A) 1.00 m
(B) 0.80 m
(C) 1.20 m
(D) 1.60 m

93. In Q92, if the force P is applied at height h = 1.2 m, the crate will:

(A) Tip over its front edge before sliding
(B) Slide smoothly without tipping
(C) Remain statically locked
(D) Lift vertically

94. A locomotive of mass 100 metric tons has 6 driving wheels. If mu_s = 0.25 between steel rails and wheels, the maximum tractive force without wheel spin is:

(A) 245.25 kN
(B) 981.00 kN
(C) 150.00 kN
(D) 60.00 kN

95. A band brake absorbs torque of 400 N*m on a 400 mm diameter drum. If mu = 0.25 and lap angle is 200 deg, the tight tension T1 is:

(A) 3424 N
(B) 2000 N
(C) 1424 N
(D) 5000 N

96. In Q95, the slack tension T2 in the band brake is:

(A) 1424 N
(B) 3424 N
(C) 2000 N
(D) 800 N

97. A heavy block of weight W is supported on a double wedge system with tapers alpha1 = 5 deg and alpha2 = 5 deg. Total mechanical efficiency is:

(A) 38.5%
(B) 50.0%
(C) 75.0%
(D) 25.0%

98. A cone clutch transmits 25 kW at 1500 rpm. The semi-cone angle alpha = 12.5 deg, mean radius r_m = 150 mm, and mu = 0.25. The axial engaging force W_a is:

(A) 1103 N
(B) 2500 N
(C) 550 N
(D) 1800 N

99. A flat pivot bearing of shaft radius R = 100 mm carries an axial thrust W = 50 kN with mu = 0.05. Under the Uniform Pressure theory, friction torque is:

(A) 166.7 N*m
(B) 125.0 N*m
(C) 250.0 N*m
(D) 83.3 N*m

100. For the pivot bearing in Q99 under the Uniform Wear theory, the friction torque is:

(A) 125.0 N*m
(B) 166.7 N*m
(C) 200.0 N*m
(D) 100.0 N*m

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State Coulomb's Laws of Dry Friction and explain the physical distinction between static friction, limiting friction, and kinetic friction.

Q2. Define the Angle of Friction, Angle of Repose, and Cone of Friction, and prove that the Angle of Repose is identically equal to the Angle of Limiting Friction.

Q3. A 500 N block rests on a rough horizontal surface with mu_s = 0.35 and mu_k = 0.30. A pulling force P inclined at 25 degrees above the horizontal is applied to the block. Calculate the magnitude of P required to (1) initiate sliding, (2) maintain uniform steady motion once moving.

Q4. A uniform ladder AB of length L = 6 m and weight W = 300 N rests against a rough vertical wall at B (mu_w = 0.25) with its lower end A on a rough horizontal floor (mu_f = 0.35). Determine the minimum angle of inclination theta with the horizontal for stable static equilibrium.

Q5. Derive the Capstan belt friction equation T1 / T2 = exp(mu * theta) for a flat belt wrapped around a fixed rough cylindrical drum.

Q6. A wedge of angle 15 degrees is used to raise a vertical load of W = 20 kN guided vertically in smooth channels. The coefficient of friction at all contacting interfaces (wedge-load, wedge-floor) is mu = 0.25. Calculate the horizontal force P required to drive the wedge.

Q7. A square-threaded screw jack has a mean diameter d_m = 50 mm and pitch p = 10 mm. The coefficient of thread friction is mu = 0.15. If a load of W = 25 kN is to be raised, calculate: (1) Helix angle, (2) Tangential effort at mean diameter, (3) Torque required, (4) Efficiency, (5) Check

Q8. A flat belt drive transmits power between two pulleys. The angle of wrap on the smaller pulley is theta = 160 degrees and mu = 0.30. If the maximum permissible belt tension is 2.5 kN, determine the maximum power that can be transmitted at a belt speed of 15 m/s (neglecting centrifugal

Q9. A uniform rectangular block of weight W = 800 N, width b = 400 mm, and height h = 800 mm rests on a rough horizontal surface with mu_s = 0.30. A horizontal pushing force P is applied at height y from the base. Determine the critical height y_crit where sliding and tipping occur

Q10. Two blocks A (weight 200 N) and B (weight 300 N) are connected by a light inextensible cord passing over a smooth pulley. Block A rests on a horizontal plane (mu = 0.25) and block B rests on a 30-degree incline (mu = 0.20). Determine the acceleration of the system and cable tension.

Q11. Derive the centrifugal tension expression T_c = m*v^2 in a moving belt and prove that maximum power occurs when T_c = T_max / 3.

Q12. Explain the mechanics of rolling resistance and derive the rolling resistance force F_r = (c_r / R) * W.

Q13. A flat belt transmits power between two pulleys with lap angle 165 deg and mu = 0.28. If allowable tension is 1500 N, find tight and slack tensions.

Q14. Analyze the differential band brake mechanism and derive the condition for self-locking.

Q15. A 200 kg block of dimensions 0.6 m width x 1.2 m height rests on a rough floor (mu_s = 0.35). Determine if horizontal force P causes sliding or toppling.

Q16. Derive the torque transmitted by a truncated conical pivot bearing under the uniform pressure and uniform wear assumptions.

Q17. Explain the operation of V-belt drives and prove why V-belts transmit greater power than flat belts for identical tension.

Q18. Calculate the holding power of a marine capstan wrapping 2.5 turns with mu = 0.35 when slack tension is 100 N.

Q19. A 1000 N block is supported on an incline with angle 25 deg. mu_s = 0.30. Find range of horizontal force P for equilibrium.

Q20. Explain disk clutch mechanics and derive the friction torque for single-plate and multi-plate clutches.

Q21. Calculate the braking torque of an internal expanding shoe brake with drum radius 150 mm and normal force 2 kN.

Q22. Prove that for an automobile negotiating a flat unbanked curved road of radius R, the maximum safe speed is v_max = sqrt(mu * g * R).

Q23. Derive the maximum height h from the ground where a force P can be applied to a uniform crate on an incline before it tips.

Q24. A screw jack with pitch 8 mm and mean diameter 40 mm lowers a 15 kN load. mu = 0.15. Calculate the torque to lower.

Q25. Discuss the role of friction in civil engineering foundation retaining walls, earth pressure, and Rankine's theory.

Q26. Calculate the minimum coefficient of static friction to prevent a block from sliding on a 35-degree incline under gravity.

Q27. Explain the stick-slip friction phenomenon and its engineering implications in machine tool slideways and earthquakes.

Q28. A belt drive transmits 15 kW between pulleys running at 20 m/s. If T1 = 2*T2, find tensions and belt cross-section if allowable stress is 2 MPa.

Q29. Explain the self-energizing and self-locking characteristics of drum brakes.

Q30. Summarize the critical engineering guidelines for design against friction failure in civil and mechanical engineering.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

strictly
inside the cone, the body remains statically stable; if R lies on the conical surface, motion impends.

2. Mathematical Proof that Angle of Repose Equals Angle of Friction (theta_r = phi_s):
Consider a block of weight W resting on a rough plane inclined at angle theta to the horizontal.
Resolve the gravitational body weight W into orthogonal components:
W_parallel = W * sin(theta) (acting down the plane, driving shear motion)
W_perpendicular = W * cos(theta) (acting normal to the plane, compressing surfaces)
From equilibrium perpendicular to the incline:
Sigma F_perp = 0 => N - W * cos(theta) = 0 => N = W * cos(theta)
From equilibrium along the incline at impending downward slip (theta = theta_r):
Sigma F_parallel = 0 => W * sin(theta_r) - F_lim = 0 => F_lim = W * sin(theta_r)
By Coulomb's friction law, F_lim = mu_s * N. Substituting for F_lim and N:
W * sin(theta_r) = mu_s * [W * cos(theta_r)]
Dividing both sides by W * cos(theta_r):
sin(theta_r) / cos(theta_r) = mu_s => tan(theta_r) = mu_s
Since by definition tan(phi_s) = mu_s, it follows directly that:
tan(theta_r) = tan(phi_s) => theta_r = phi_s.
Thus, the angle of repose is proved to be identically equal to the angle of limiting friction.

Solution to Question 3:

P * cos(25 deg) - mu_k * N = 0
0.9063 * P - 0.30 * [500 - 0.4226 * P] = 0
0.9063 * P - 150.0 + 0.1268 * P = 0
1.0331 * P = 150.0
P_maintain = 150.0 / 1.0331 = 145.19 N.

5. Final Results:
- Force required to initiate motion: P_init = 166.00 N.
- Force required to maintain steady sliding: P_maintain = 145.19 N.

>> Solution to Question 4: A uniform ladder AB of length L = 6 m and weight W = 300 N rests against a rough vertical wall at B (mu_w = 0.25) with its lower end A

degrees.

Angle of friction: phi = atan(mu) = atan(0.25) = 14.04 degrees.

2. Analysis of the Vertical Load Block:
Forces acting on the raised block:
- Downward load: W = 20 kN.
- Horizontal reaction from smooth vertical guide: R_guide.
- Total contact reaction R1 from inclined face of the wedge.
Since the wedge moves horizontally to the right, the block slides up the wedge incline.
Therefore, reaction R1 is inclined at (alpha + phi) to the vertical:
theta_1 = alpha + phi = 15.0 deg + 14.04 deg = 29.04 degrees to the vertical.
Applying vertical equilibrium to the block (Sigma Fy = 0):
R1 * cos(alpha + phi) - W = 0
R1 * cos(29.04 deg) = 20 kN
R1 = 20 / cos(29.04 deg) = 20 / 0.8743 = 22.88 kN.

3. Analysis of the Horizontal Wedge:
Forces acting on the driving wedge:
- Horizontal driving force: P directed to the right.
- Normal and friction reaction from bottom floor: R2.
Since wedge moves right, friction on floor acts to the left.
Hence, reaction R2 is inclined at angle phi = 14.04 deg to the vertical.
- Action force from load block: R1 = 22.88 kN acting downward and left at 29.04 deg to vertical.
Applying horizontal and vertical equilibrium to the wedge:
Sigma Fy = 0 => R2 * cos(phi) - R1 * cos(alpha + phi) = 0
R2 * cos(14.04 deg) = 22.88 * cos(29.04 deg) = 20.0 kN
R2 = 20.0 / cos(14.04 deg) = 20.0 / 0.9701 = 20.62 kN.
Applying horizontal equilibrium (Sigma Fx = 0):
P - R2 * sin(phi) - R1 * sin(alpha + phi) = 0
P = [20.62 * sin(14.04 deg)] + [22.88 * sin(29.04 deg)]
P = (20.62 * 0.2425) + (22.88 * 0.4854) = 5.00 + 11.11 = 16.11 kN.

4. Conclusion:
The horizontal driving effort required to raise the 20 kN foundation load is P = 16.11 kN.

>>Solution to Question 7: A square-threaded screw jack has a mean diameter d_m = 50 mm and pitch p = 10 mm. The coefficient of thread friction is mu = 0.15.

P = 25,000 N * 0.2157 = 5392.5 N = 5.39 kN.

3. Torque Required to Raise Load (T_raise):
T_raise = P * (d_m / 2) = 5392.5 N * 0.025 m = 134.81 N-m.

4. Mechanical Efficiency of Screw Jack (eta):
Efficiency is the ratio of ideal torque without friction to actual torque:
eta = tan(alpha) / tan(alpha + phi)
eta = 0.06366 / 0.2157 = 0.2951 = 29.51%.

5. Self-Locking Condition Check:
Condition for self-locking: alpha <= phi.
Here, alpha = 3.64 deg and phi = 8.53 deg. Since alpha < phi (and efficiency < 50%), the screw jack is strictly SELF-LOCKING. The load will not lower under its own weight when operating torque is removed.

(b) Impending Tipping occurs when the normal reaction shifts completely to the pivot edge (front corner).
Taking moments about the pivot edge:
Overturning moment = P * y; Restoring moment = W * (b / 2).
At impending tipping: P_tip * y = W * (b / 2) => P_tip = W * (b / 2*y).

2. Simultaneous Sliding and Tipping Condition:
Simultaneous impending sliding and tipping occurs when P_slip = P_tip:
mu_s * W = W * (b / 2*y_crit)
Cancelling weight W from both sides:
mu_s = b / (2 * y_crit)
y_crit = b / (2 * mu_s)

3. Numerical Calculation:
Given width b = 400 mm and mu_s = 0.30:
y_crit = 400 mm / (2 * 0.30) = 400 / 0.60 = 666.67 mm.
Magnitude of Force at Impending Transition:
P = mu_s * W = 0.30 * 800 N = 240.0 N.

4. Physical Significance:
- If y < 666.67 mm: The block will SLIDE before it can tip.
- If y > 666.67 mm: The block will TIP over before it can slide.
- At y = 666.67 mm: Both impending sliding and impending tipping occur at exactly P = 240 N.

>> Solution to Question 10

Two blocks A (weight 200 N) and B (weight 300 N) are connected by a light inextensible cord passing over a smooth pulley. Block A res

(T - 50.0) + (98.04 - T) = (20.39 + 30.58) * a
48.04 = 50.97 * a
a = 48.04 / 50.97 = 0.943 m/s^2.
Substitute 'a' into Equation 1 to find cable tension T:
T = 50.0 + (20.39 * 0.943) = 50.0 + 19.23 = 69.23 N.

4. Conclusion:
System acceleration is a = 0.943 m/s^2; Cable tension is T = 69.23 N.

>> Solution to Question 11: Derive the mechanical efficiency at maximum power for a screw jack. Solution: Efficiency at maximum power occurs when T_c = T_max / 3.

Governing criteria involve Coulomb's dry friction law, non-concurrent moment equilibrium, and normal/tangential coordinate projections at contacting interfaces.

2. Mathematical Formulation & Step-by-Step Analysis:
Mechanics of Rolling Resistance:
1. Microscopic Asymmetry:
When a hard cylinder or wheel rolls over a deformable ground plane under normal load W, the contact surface indents. Due to internal material hysteresis and elastic recovery lag, the pressure distribution is non-symmetric, shifting forward by distance c_r.
2. Moment Equilibrium:
The resultant normal ground reaction N = W is shifted forward by distance c_r from the wheel center line.
To maintain steady rolling at constant velocity, driving horizontal force F_r applied at center O must overcome the restoring couple:
F_r * R = W * c_r => F_r = (c_r / R) * W.
3. Coefficient of Rolling Resistance:
c_r has units of length (mm or m), representing the forward shift of normal reaction.

3. Detailed Physical Verification & Equilibrium Check:
- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>> Solution to Question 13: A flat belt transmits power between two pulleys with lap angle 165 deg and mu = 0.28. If allowable tension is 1500 N, find tight and slac

In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>>Solution Statement: Assume the block is in equilibrium and derive the condition for self-locking (tan = 0.35). Determine if horizontal force P causes sl

Sigma M_C = 0 => P_tip * h = W * (width / 2)
P_tip * 1.2 = 1962 * 0.30 => P_tip = 588.6 / 1.2 = 490.5 N.
3. Conclusion:
Since P_tip (490.5 N) < P_slide (686.7 N), the block will TIP OVER its corner before it can slide!

3. Detailed Physical Verification & Equilibrium Check:
- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>> Solution Step 1: Identify the operation of welding as a cantilever beam under uniform load for identical assumptions.

Governing criteria involve Coulomb's dry friction law, non-concurrent moment equilibrium, and normal/tangential coordinate projections at contacting interfaces.

2. Mathematical Formulation & Step-by-Step Analysis:
V-Belt Wedge Action and Power Capacity:
1. Groove Wedging Mechanics:
A V-belt fits into a pulley groove of included angle 2*beta (groove semi-angle beta approx 18 deg).
Radial force equilibrium: 2 * R_N * sin(beta) = dN => R_N = dN / (2 * sin(beta)).
2. Enhanced Frictional Resistance:
Total friction dF = 2 * mu * R_N = 2 * mu * [dN / (2 * sin(beta))] = [mu / sin(beta)] * dN.
Effective friction coefficient: mu_prime = mu / sin(beta).
3. Power Comparison:
Since beta approx 18-20 deg, sin(beta) approx 0.31-0.34, making mu_prime approx 3 * mu!
The tension ratio T1/T2 = exp(mu_prime * theta) increases dramatically, generating vastly larger (T1 - T2) and power without slipping.

3. Detailed Physical Verification & Equilibrium Check:
- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>> **Solution to Question 6.18:** Calculate the holding power of a marine capstan wrapping 2.5 turns with mu = 0.35 when slack tension is 100 N.

>>Solution Statement Question 5 2000 kg block is connected and drive with friction 25 deg from rest 0.30. Find radius of the force P for equilibrium.

- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>>Solution Code: Question 22: Calculate the maximum safe speed for a car rounding a flat unbanked curve with radius R = 150 m and coefficient of friction mu = 0.6. v_max = sqrt(mu * g * R)

- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>>Solution Statement Question 23: The maximum height from the ground level at which a 15 kN load can be lifted using a 40 m long crane is 8 m. Calculate the moment of the load.

- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>> Solution Statement Question 26: Use the coefficient of static friction between a block and a surface and Rankine's theory.

3. Detailed Physical Verification & Equilibrium Check:
- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

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3. Cross-Sectional Area:
Area A = T_max / sigma_allow = 1500 N / (2 * 10^6 N/m^2) = 7.5 * 10^-4 m^2 = 750 mm^2.

3. Detailed Physical Verification & Equilibrium Check:
- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.

>> Solution to Question 30: Analyze self-energizing and self-locking mechanisms for design of friction

1. Maximize Friction Where Essential:
- Apply high-strength friction-grip (HSFG) bolts in steel structures.
- Maintain safety factor >= 1.5 against base sliding in dams and retaining walls.
- Utilize grooved V-belts and textured road surfaces for tire traction.
2. Minimize Friction Where Detrimental:
- Provide fluid-film hydrodynamic lubrication in journal bearings.
- Install elastomeric or PTFE sliding expansion bearings on bridge piers.
3. Ensure Self-Locking in Lifting Equipment:
- Design screw jacks with alpha < phi (efficiency < 50%) to prevent catastrophic back-driving drops.

3. Detailed Physical Verification & Equilibrium Check:
- Normal contact forces must satisfy Sigma F_n = 0 across all boundary contact interfaces.
- Frictional forces must not exceed limiting value F <= mu * N for stable static equilibrium.
- Moment balance Sigma M = 0 ensures no rotational tipping or rocking instability occurs.

4. Civil & Structural Engineering Implications:
In foundation retaining walls, dam bases, and structural bolted connections, an adequate factor of safety (FS >= 1.5) against sliding is mandatory under severe seismic and wind loads.
